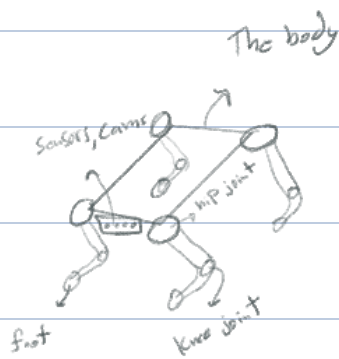




- material: plastic

- degrees of freedom:

for each leg $\rightarrow$ hip joint = 1 degree of freedom

$\rightarrow$ knee joint = 1 degree of freedom

So, 4 legs $\times$ 2 degrees of freedom = 8 total degrees of freedom

- motor selection: servo motor, because it gives you precise control of the angle of each joint and it's the best for robotic legs, total 8 servo motor, one for each joint

- Calculating the initial torque of a joint:

if a joint carry: mass = 1 kg

Distance = 0.3 m

strength: $F = m \times g$

$$= 1 \times 9.81$$

$$= 9.81$$

determination: $T = F \times r$

$$= 9.81 \times 0.3$$

$$= 2.94 \text{ N.m}$$

$\neq$ so we need an engine with a torque at least 2.94 N.m, but we recommend a bit higher due to different reasons such as friction

- stability and center of gravity:

- battery is placed in the middle of the body

- center of gravity is low also close to the middle

- legs are distributed in the corners increase stability and reduce over turning

- walking method :-

mutual steps :- front left + right back

Then

right front + left rear

good stability while moving

- mechanical Problems expected :

- friction in the joint

- Choosing an engine with not enough torque

- high center of gravity This will cause over turning

- failure to align the joints

- weak or heavy structure